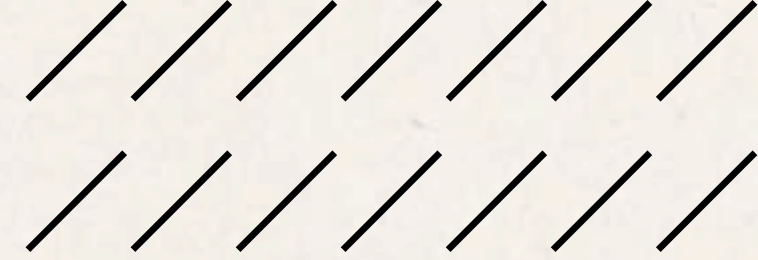


FTM315 FINAL UNIVERSAL DESIGN

Collin Stockmoe

The Objective

- 01** Apply fundamental patternmaking and technical design principles to develop a cohesive, well-constructed, fully patterned garment
- 02** Follow **NC State's Principles of Universal Design** to create an intuitive, functional, and accommodating design solution.
- 03** Work within the guides of an assigned fit model, incorporating appropriate ease, balance, and grainline alignment to achieve accurate fit and proportion.



What is Universal Design?

Developed by **NC State's Center for Universal Design**, the *Principles of Universal Design* serve as a foundation for creating products and experiences that are intuitive, functional, and accessible to a broad range of users. In apparel, these principles encourage thoughtful consideration of fit, usability, and comfort—ensuring garments accommodate diverse needs without compromising design intent.

Core Principles

- 01 Equitable Use** - The design is useful and marketable to people with diverse abilities
- 02 Flexibility in Use** - The design provides choice in methods of use, adaptability to the user's pace, and facilitate the user's accuracy and precision
- 03 Simple & Intuitive Use** - Use of the design is easy to understand, eliminating unnecessary complexity and allows consistent user expectations and intuition.
- 04 Perceptible Information** - The design communicates necessary information effectively to the user, regardless of ambient conditions or user's sensory abilities
- 05 Tolerance for Error** - The design minimizes hazards and the adverse consequences of accidental or unintended actions
- 06 Low Physical Effort** - The design can be used efficiently and comfortably with a minimum of fatigue
- 07 Size and Space for Approach and Use** - Appropriate size and space is provided for approach, reach, manipulation, and use regardless of user's body size, posture, or mobility.



Design Concept

For my garment, I drew inspiration from the utility and practicality of vintage military uniforms, specifically the storage-focused design of the M-65 field pant and the functional flexibility of traditional U.S. Navy sailor trousers.

Key Solutions

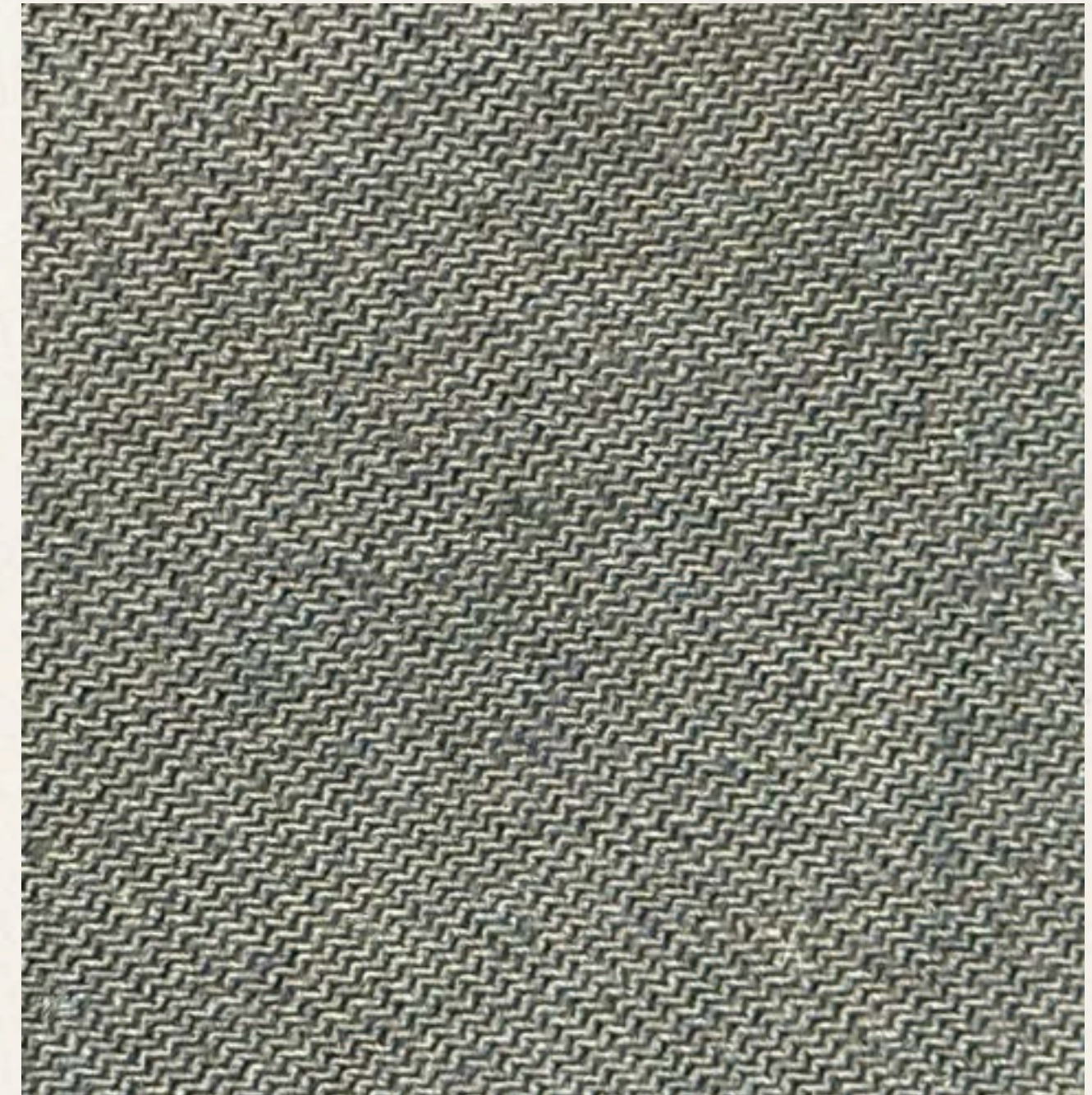
A **high-rise** waist provides secure coverage and comfort in movement, while **generous ease** throughout the garment allows for flexibility and accessibility across body types. The addition of an **adjustable back belt tab with button closures** introduces a customizable fit, accommodating variations in waist size and preference without compromising structure. **Military-inspired pockets** enhance both storage and function. Lastly, a **heavyweight 14oz denim** adds durability and structure.

Fabric Swatch

Fabric - 14oz right-hand 3×1 twill denim, 425GSM

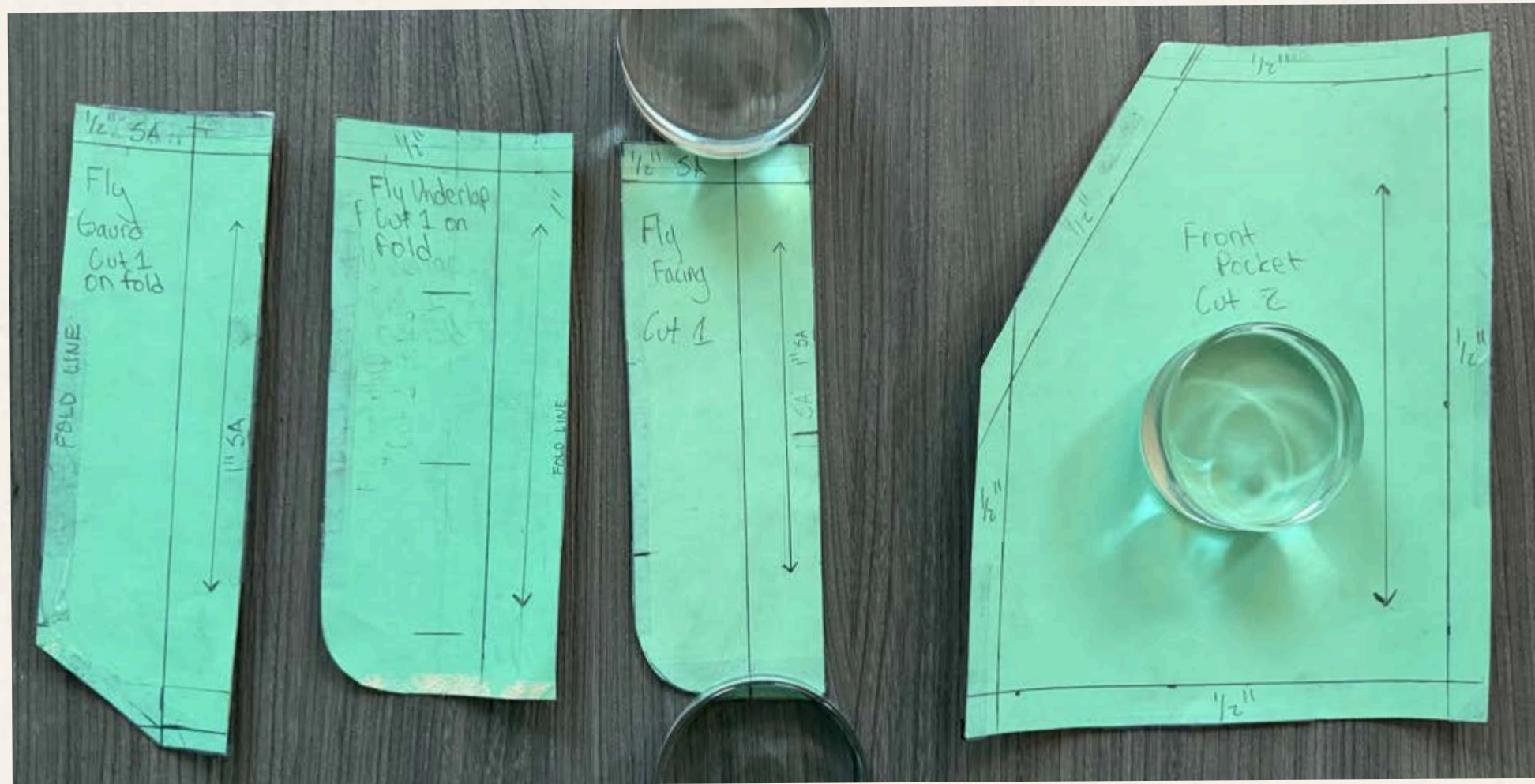
Source - Milled in North Carolina

Description - Classic 3×1 right-hand twill structure providing durability, structure, and a defined surface texture. The heavier weight supports longevity while maintaining flexibility and comfort through wear. Milled by *Cone Denim*, a heritage North Carolina mill recognized for its high-quality, American-made selvedge and workwear fabrics.



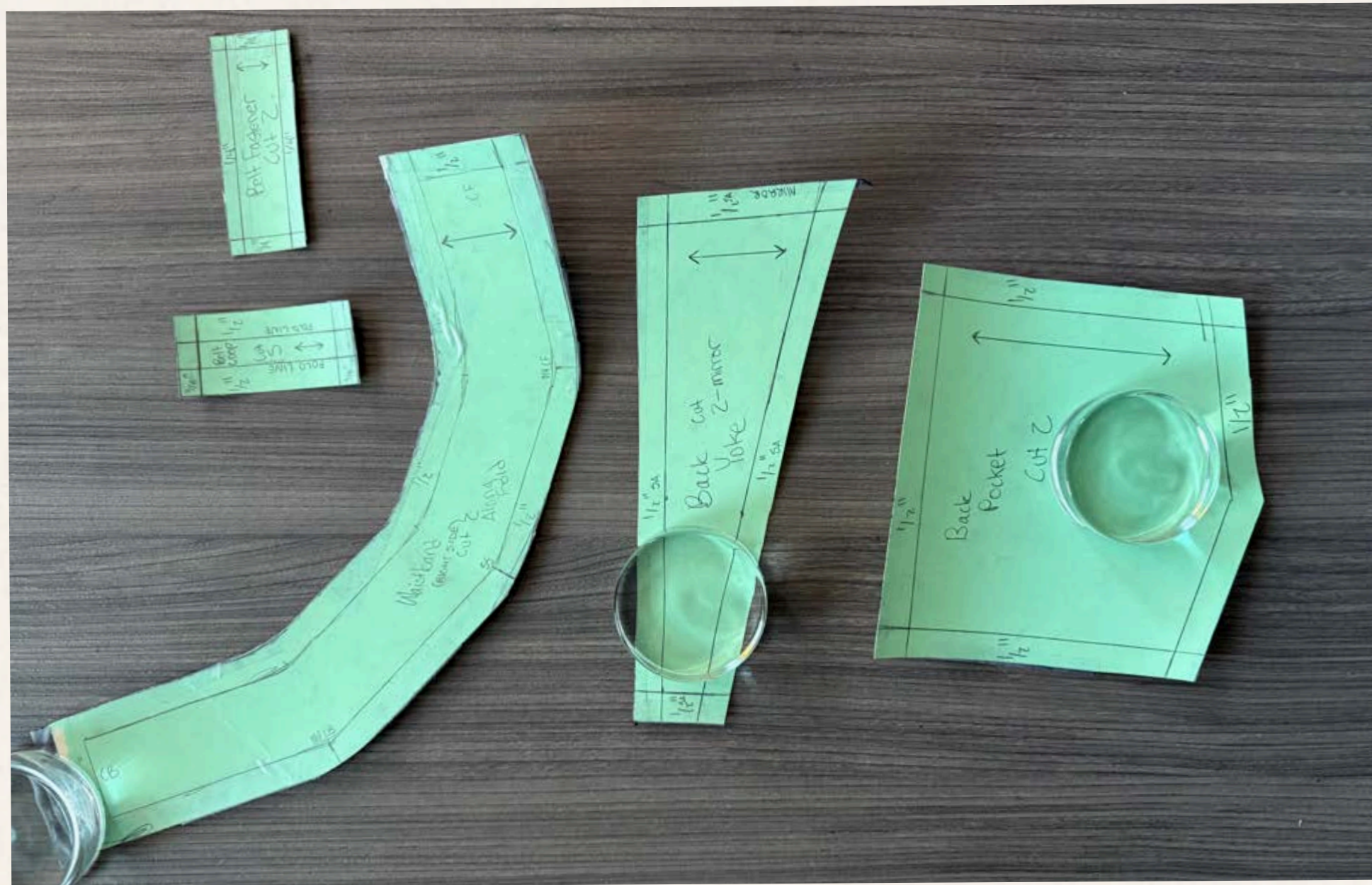
Patternmaking

Centerfront + Front Pockets + Fly



Patternmaking

Centerback + Waistband + Belt Loops + Belt Fastener + Back Pockets + Back Yoke



The Process

For the CF & CB patterns, I began by drafting a base pant pattern using principles outlined in Helen Joseph Armstrong's Patternmaking for Fashion Design, incorporating the foundational measurements of my fit model. From there, I incorporated 2.5" of ease through the hip and thigh to enhance comfort and range of movement, while flaring the leg opening to create a balanced, wide-leg silhouette that maintains proportion and fluidity.

The functional elements, including pockets, fly, belt loops, waistband, and the back button-tab, were drafted directly into the pattern to ensure precise placement and seam alignment. To refine fit at the waist, I incorporated a curved waistband that contours to the body while maintaining adjustability through the back buttoned belt tab, allowing for a more customizable fit.

Finished Garment



Fit Notes

Overall, the fit through the thigh and leg carried the intended balance of structure and ease, allowing for comfortable movement without distortion. The downward angle of the back yoke complemented the high rise effectively in how it shifted ease throughout the leg, and the curved waistband refined the waist contour for a clean, ergonomic fit.

For future iterations, I would extend the fly by $\frac{1}{4}$ inch toward the outseam to better accommodate the button closure and improve front overlap. Additionally, I would tighten the waist and hip area by adjusting the crotch curve inward approximately $\frac{1}{4}$ inch to achieve a closer, more tailored fit through the seat.